



Crop Batch Management & Analytics Portal for Agriculture & AgriTech

Review 1 | Team 16 | Department of Computer Science & Engineering



Team Details

Team Members

Anikha Padala

Roll No: **2520030613**

Mannepalli Jashwanth

Roll No: **2520030590**

Project Guide

[To be added]

Department

Computer Science & Engineering

Review

Review 1 — Team 16

Abstract



Agriculture generates vast amounts of data across crops, farmers, fields, and cultivation activities — information that becomes difficult to track and analyze when managed manually. The **Crop Batch Management & Analytics Portal** addresses this by offering a centralized, database-driven system to record and monitor farmers, fields, crops, and crop batches, along with key activities like irrigation, fertilization, pest/disease tracking, and harvesting.

The system enables end-to-end tracking of each crop batch's lifecycle, supporting easy addition, updates, and retrieval of records. It also delivers analytical insights — including crop-wise production, batch status, yield trends, and resource utilization — to support data-driven decision-making in agriculture.

Introduction

Data-Intensive Domain

Agriculture requires management of information across farmers, fields, crops, and cultivation activities at scale.

Manual Inefficiencies

Manual record-keeping leads to inefficiencies, data inconsistencies, and difficulty in tracking crop batches.

Centralized Solution Needed

Organizations need a centralized, database-driven solution for managing agricultural data effectively.

Analytics for Decision-Making

Reporting and analytics capabilities are essential for informed decision-making in agriculture and AgriTech.



Existing vs. Proposed System

EXISTING SYSTEM

Current Challenges

Manual Record-Keeping

Spreadsheets or paper-based systems with scattered data across multiple sources.

No Centralized Repository

Difficulty tracking crop batch lifecycle; inconsistent data formats and human error.

Limited Analytics

Time-consuming data retrieval and minimal ability to generate meaningful reports.

PROPOSED SYSTEM

Our Solution



Centralized Database

Structured records for all agricultural entities in one platform.



Real-Time Analytics

Insights on production, yield, batch status, and resource utilization.



Lifecycle Tracking

Monitor every crop batch from planting through harvest efficiently.

Problem Statement



The Core Challenge

Agricultural organizations today face a critical gap between the volume of data they generate and their ability to manage it effectively. Manual processes introduce errors, fragment data across systems, and leave decision-makers without the insights they need. There is a clear and urgent need for a structured, centralized solution.

⚠ Data fragmentation and manual errors directly impact crop yield decisions and resource planning.



Solution



Structured Data Models

Implement models for farmers, fields, crops, and crop batches.



User-Friendly Interface

Easy add, update, and retrieval of agricultural records.



Analytical Processing

Generate insights, reports, and decision-support features.



Lifecycle Monitoring

Track crop batches with all associated cultivation activities.

Modules



Farmer & Field Management

Register and manage farmer profiles and field information with structured records.



Crop Batch Management

Create, track, and manage crop batches throughout their complete lifecycle.



Cultivation Activity Tracking

Record irrigation, fertilizer usage, pest observations, and harvesting activities.



Data Management

Add, update, retrieve, and delete agricultural records with full CRUD operations.



Analytics & Reporting

Generate insights on crop-wise production, batch status, yield, and resource utilization.



User Management

Authentication and role-based access control for secure system usage.

Software Requirements



MySQL

Database design, data manipulation, and querying



HTML / CSS / JavaScript

User interface development and frontend interactions



Python

Backend application development and business logic



Visual Studio Code

Development environment with Git/GitHub for version control

Conclusion

Comprehensive Solution

The portal provides a complete solution for managing agricultural data across the entire crop lifecycle.

Data Consistency

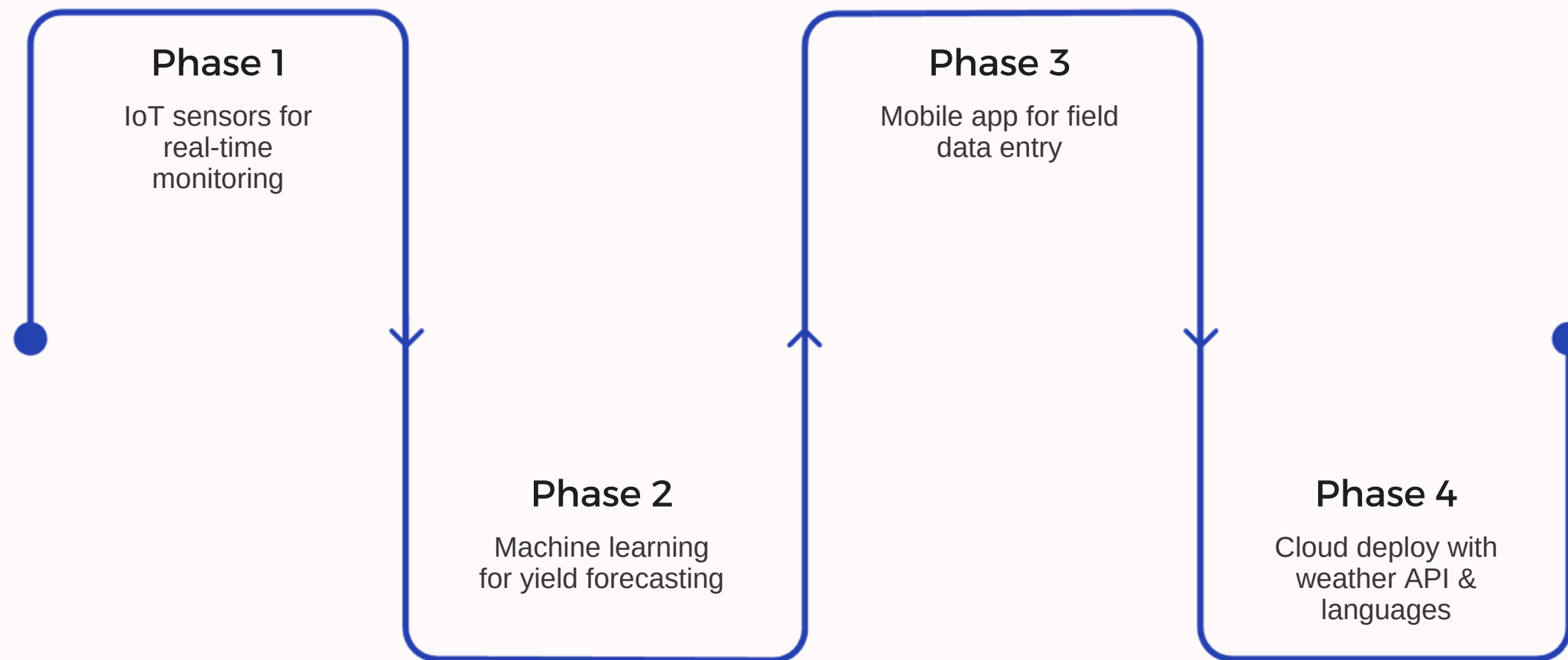
Centralized database approach ensures data consistency, accessibility, and efficient retrieval at all times.

Better Decisions

Analytical capabilities enable better decision-making and actionable insights for agriculture and AgriTech.



Future Scope



Expanding the Platform

- **IoT Integration:** Real-time crop monitoring via sensor networks
- **Machine Learning:** Predictive insights and yield forecasting
- **Mobile Application:** Field-level data entry and monitoring on the go
- **Weather APIs:** Climate-aware recommendations for farmers
- **Multi-Language Support:** Broader accessibility across regions
- **Cloud Deployment:** Scalability and remote accessibility

□ The portal is designed as a scalable foundation — each future enhancement builds on the centralized data architecture established in this phase.



Thank You

We appreciate your time and att